

- 5 A steel ball on the end of a thin string oscillates with small oscillations, as shown in Fig. 5.1.

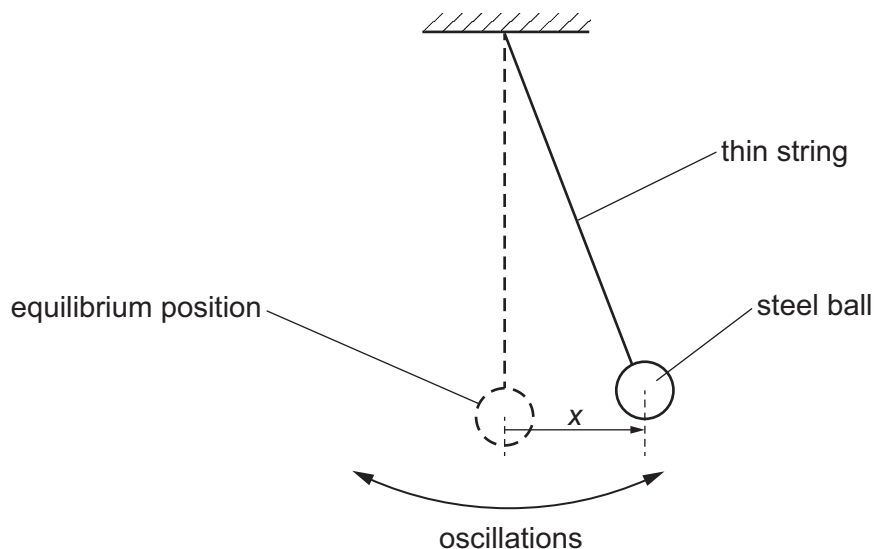


Fig. 5.1 (not to scale)

The displacement of the centre of the ball from its equilibrium position is x .

- (a) Fig. 5.2 shows the variation with x of the acceleration a of the ball.

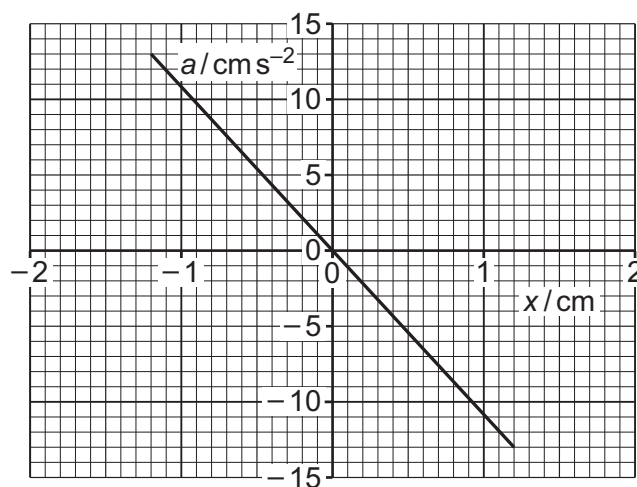


Fig. 5.2

- (i) Explain how Fig. 5.2 shows that the oscillations of the ball are simple harmonic.

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